

757. Set Intersection Size At Least Two

Difficulty: Hard

Link: <https://leetcode.com/problems/set-intersection-size-at-least-two/description/?envType=daily-question&envId=2025-11-20>

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Problem:

You are given a 2D integer array `intervals` where `intervals[i] = [starti, endi]` represents all the integers from `starti` to `endi` inclusively.

A **containing set** is an array `nums` where each interval from `intervals` has **at least two** integers in `nums`.

- For example, if `intervals = [[1,3], [3,7], [8,9]]`, then `[1,2,4,7,8,9]` and `[2,3,4,8,9]` are **containing sets**.

Return *the minimum possible size of a containing set*.

Example 1:

Input: `intervals = [[1,3],[3,7],[8,9]]`

Output: 5

Explanation: let `nums = [2, 3, 4, 8, 9]`.

It can be shown that there cannot be any containing array of size 4.

Example 2:

Input: intervals = [[1,3],[1,4],[2,5],[3,5]]

Output: 3

Explanation: let nums = [2, 3, 4].

It can be shown that there cannot be any containing array of size 2.

Example 3:

Input: intervals = [[1,2],[2,3],[2,4],[4,5]]

Output: 5

Explanation: let nums = [1, 2, 3, 4, 5].

It can be shown that there cannot be any containing array of size 4.

Constraints:

- $1 \leq \text{intervals.length} \leq 3000$
- $\text{intervals}[i].\text{length} == 2$
- $0 \leq \text{start}_i < \text{end}_i \leq 10^8$